

ENG 101
Lesson -10
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Mainframes:

Large computer systems, or **mainframes**, as they are referred to in the field of computer science, are those Computer systems found in **computer installations** processing immense amounts of data. These powerful computers make use of very high-speed main memories into which data and programs to be dealt with are transferred for rapid access. These powerful machines have a larger repertoire of more complex instructions which can be executed more quickly. Whereas smaller computers may take several steps to perform a particular operation, a larger machine may accomplish the same thing with one instruction.

These computers can be of two types: **digital** or **analog**. The digital computer or general-purpose computer as it is often known, makes up about 90 percent of the large computers now in use. It gets its name because the data that are presented to it are made up of a code consisting of **digits**—single-character numbers. The digital computer is like a gigantic cash register in that it can do calculations in steps, one after another at tremendous speed and with great accuracy.

Digital computer **programming** is by far the most commonly used in electronic **data processing** for business or statistical purposes. The analog computer works something like a car speedometer, in that it continuously works out calculations. It is used essentially for problems involving measurements. It can simulate, or imitate different measurements by electronic means. Both of these computer types – the digital and the analog – are made up of electronic components that may require a large room to accommodate them. At present, the digital computer is capable of doing anything the analog once did. Moreover, it is easier to program and cheaper to operate. A new type of scientific computer system called the hybrid computer has now been produced that combines the two types into one.

Really powerful computers continue to be bulky and require special provision for their housing, refrigeration systems, air filtration and power supplies. This is because much more space is taken up by the input/output devices – the magnetic tape and disk units and other peripheral equipment – than by the electronic components that do not make up the bulk of the machine in a powerful installation. The power consumption of these machines is also quite high, not to mention the price that runs into thousands of dollars.

The future will bring great developments in the mechanical devices associated with computer systems. For a long time these have been the weak link from the point of view of both efficiency and reliability.

Exercises :

1. Main idea

Which statement best expresses the main idea of the text? Why did you eliminate the other choices?

- ☐ 1. Hybrid computers are a combination of digital and analog computers.
- ☐ 2. Digital computers are used more than any other type of computer.
- ☐

3. There are three types of mainframes.

☐

4. Analog computers can do more varied work than digital or hybrid computers.

2. Understanding the passage:

Decided whether the following sentences are true or false (T/F) by referring to the information in the text. Then make the necessary changes so that the false statements become true.

☐☐

1. A mainframe is the type of computer that can sit on top of a desk.

☐☐

2. Mainframes are very powerful and can execute jobs very rapidly and easily.

☐☐

3. Digital computers are used more than analog computers.

☐☐

therefore 4. The analog computer is far smaller than a digital computer and occupies very little space.

☐☐

5. The hybrid computer is a combination of both the digital and the analog computer.

☐☐

6. The analog computer does its calculations one step at a time.

☐☐

7. The digital computer continuously works out calculations.

☐☐

8. Mainframes are huge powerful machines whose peripheral equipments takes up a lot of space.

☐☐

9. Mainframes are expensive to buy and to operate.

☐☐

10. Mainframe technology has reached the end of the road. No further development is needed.

3. Locating information :

Find the passages in the text where the following ideas are expressed. Give the line references.

- 1. Smaller computers may take longer to perform an operation.
-2. More technological development is necessary in the mechanical devices associated with computer systems.
-3. Mainframes can operate quickly and execute more complex instruments.
-4. The hybrid computer is a combination of both digital and analog computers.
-5. Digital computers are used more than analog computers.
-6. Mainframes are large powerful computers.
-7. An analog computer is comparable to a car speedometer in the way it operates.
- 8. Digital computers do calculations, one after another, quickly and exactly.

4. Contextual reference :

Look back at the text and find out what the words in **bold** typeface refer to.

1. **which** can be executed

(l.7)

2.2. as it is often known	(l.12).....
3. that are presented to it	(l.14).....
4. in that it can do calculations	(l.16).....
5. in that it continuously works out	(l.20).....
6. Both of these computer types	(l.23).....
7. that may require a large room	(l.24).....
8. that combines the two types	(l.29).....
9. Require special provision for their housing	(l.31).....
10. that runs into thousands of dollars	(l.37).....

5. Understanding words

Refer back to the text and find synonyms for the following words.

1. area	(l. 2)
2. Acted on	(l. 7)
3. do	(l. 9)
4. Composed of	(l. 15).....
5. Principally	(l. 21)

Now refer back to the text and find antonyms for the following words.

6. ignored	(l. 5)
7. seldom	(l. 12)
8. little	(l. 16)
9. small	(l. 30).....
10. weak	(l. 35)

6. Words forms:

First choose the appropriate form of the words to complete the sentences. Then check the differences of meaning in your dictionary.

1. permission, permit, permissible, permitted
 - a. It is usually not to smoke in a computer installation.
 - b. Computers people to use their time more effectively.
 - c. Building is usually required before starting any renovations to a building for computer department.
2. continuation, continue, continuing, continuously
 - a. If microcomputer sales to increase, it won't be long before every household has one.
 - b. Computers can do repetitive operations..... without getting bored.
 - c. There is ainterest in discovering new areas where computers can be used.
3. measurement, measure, measured, measurable
 - a. The analog computer is essentially used for problems is involving
 - b. Because computer equipment is often bulky, the area used for a computer installation must be..... out carefully.
 - c. The number of employees a computer company has can be seen as a of its success in the business world.
4. association, associate, associated
 - a. Computers are with speed and accuracy.
 - b. There are many computer..... around the world to which computer professionals belong.
 - c. Business In different cities often communicate with each other via their computers.
5. Efficiency, efficient, efficiently
 - a. Using a hand calculator to do simple mathematics is anway of working.
 - b. Computers can solve problems faster and more than humans.

c..... is important in any service industry.

7. Content review:

Try to think of a definition for each of these items before checking them in the Glossary. Then complete the following statements with the appropriate words. (Some can be used more than once.) Make sure you use the correct form, i.e. singular or plural.

mainframe	computer installation	digits
hybrid computer	code	programming
Digital	analog	

1. The system is a computer which has combined the features of both the and computer. It is used mainly in scientific research.
- 2.....computers get their name from the word.....These are single character numbers that make up the in which the data are presented to the computer for processing.
- 3..... are usually found in large
4. The most commonly used language of in the business community is

8. Focus review :

Focus C Organizing information

On a separate sheet, organize the information in Unit 6, 'Mainframes', under main idea(s) major details and minor details.